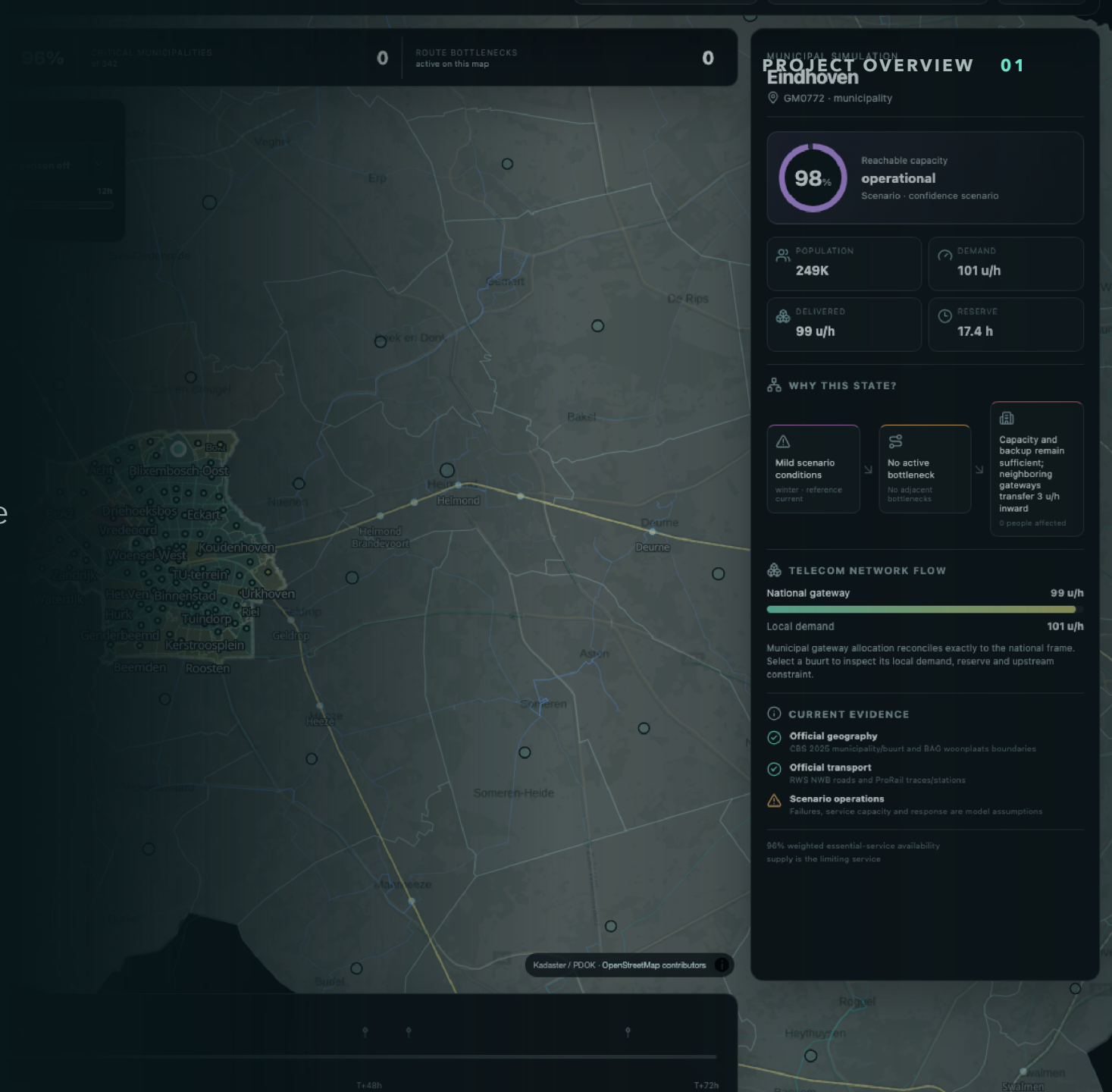
NL RESILIENCE LAB

DISASTERSIMULATOR

Explore the first 72 hours.

A transparent disaster-cascade simulator for the Netherlands.

Rik Stokmans • 4 September 2026



THE PREMISE

A disaster is not one failure.

It is a chain of dependencies, delays and constrained choices.

01 Shock

Wind, water, heat, cyber, ground motion or fire creates the initiating pressure.

02 Cascade

Power, telecom, water, mobility and supply degrade on different clocks.

03 Choice

Preparedness, backup, access and repair capacity change what happens next.

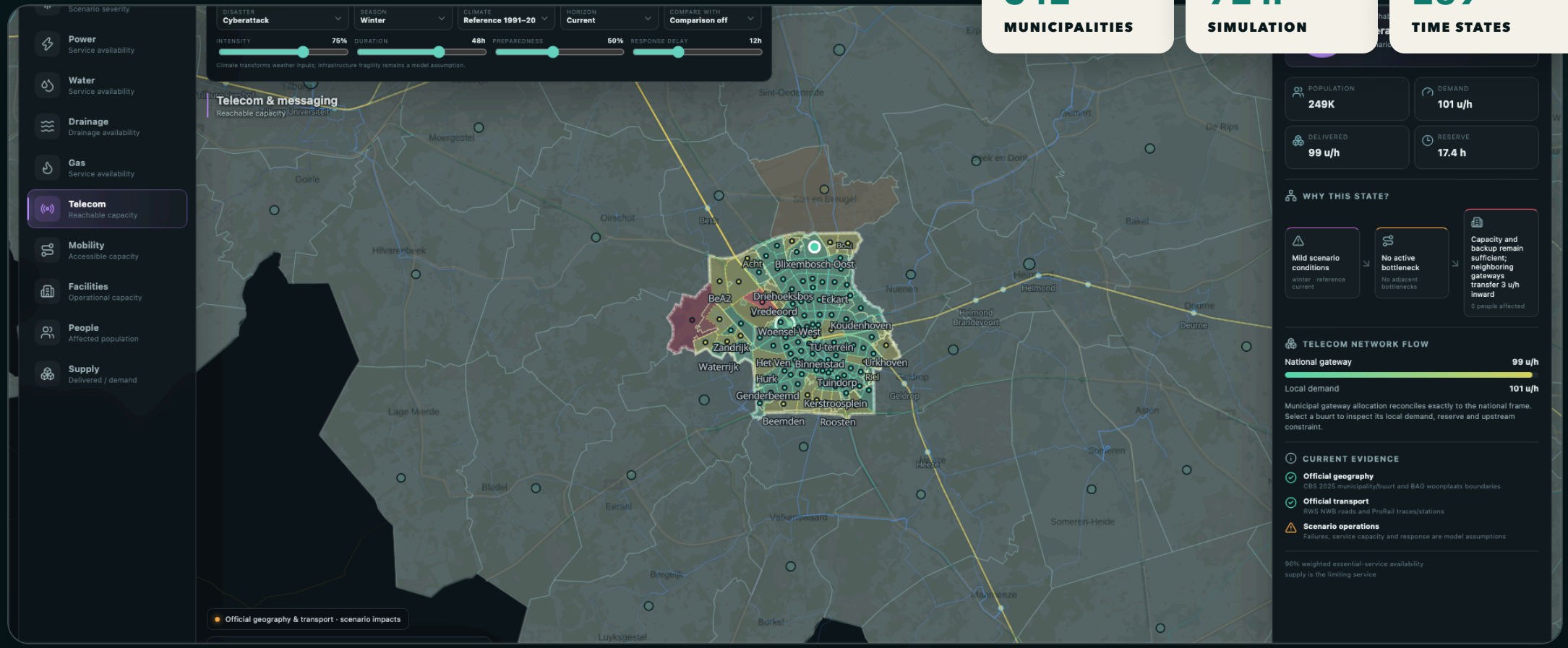
ONE COHERENT STATE

National overview. Local consequences.

342 MUNICIPALITIES

72 h SIMULATION

289 TIME STATES



The same scenario, time and service state remain coherent while the view changes.

INSIDE THE ENGINE

Six visible stages. One repeatable result.

01

Scenario

Hazard, climate, intensity, start and duration.

→ 02

Hazard

Time envelope, location and audited guards.

→ 03

Damage

Service pressure, resilience and preparedness.

→ 04

Cascade

Power and mobility constrain dependent services.

→ 05

Balance

Supported routes reduce local shortfalls.

→ 06

Explain

Status, causes, impacts and counterfactuals.

Same configuration + stable seed → same state and explanation

Repeatable does not mean forecast-certain.

THE CASCADE

Two structural hubs shape the outcome.

POWER

Power constrains services.

Backup buys time, not immunity.

Water
backup window

Telecom
battery window

Drainage
continuous need

Gas controls
partly passive

Facilities
generator support

MOBILITY

Mobility constrains supply.

Access governs road-based logistics.

Road access
NWB corridors

Logistics
route capacity

Emergency supply
delivery potential

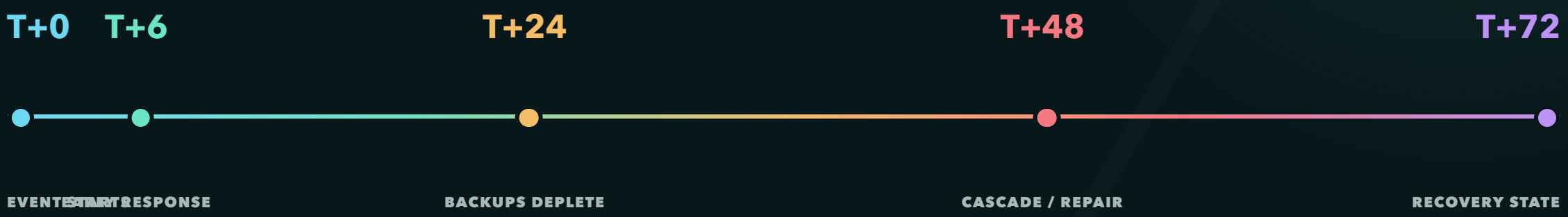
Repair
access guard

Road geometry is causal only for Mobility and explicitly modelled road logistics—not as a substitute for utility topology.

THE FIRST 72 HOURS

The answer changes with the clock.

Shock, backup endurance, cascading shortfall and restoration do not peak together.



RESPONSE DELAY
When organised restoration may begin.

BACKUP ENDURANCE
How long dependencies stay decoupled.

ACCESS GUARD
When standing water can pause repair.

PEAK DAMAGE
What may persist after forcing declines.

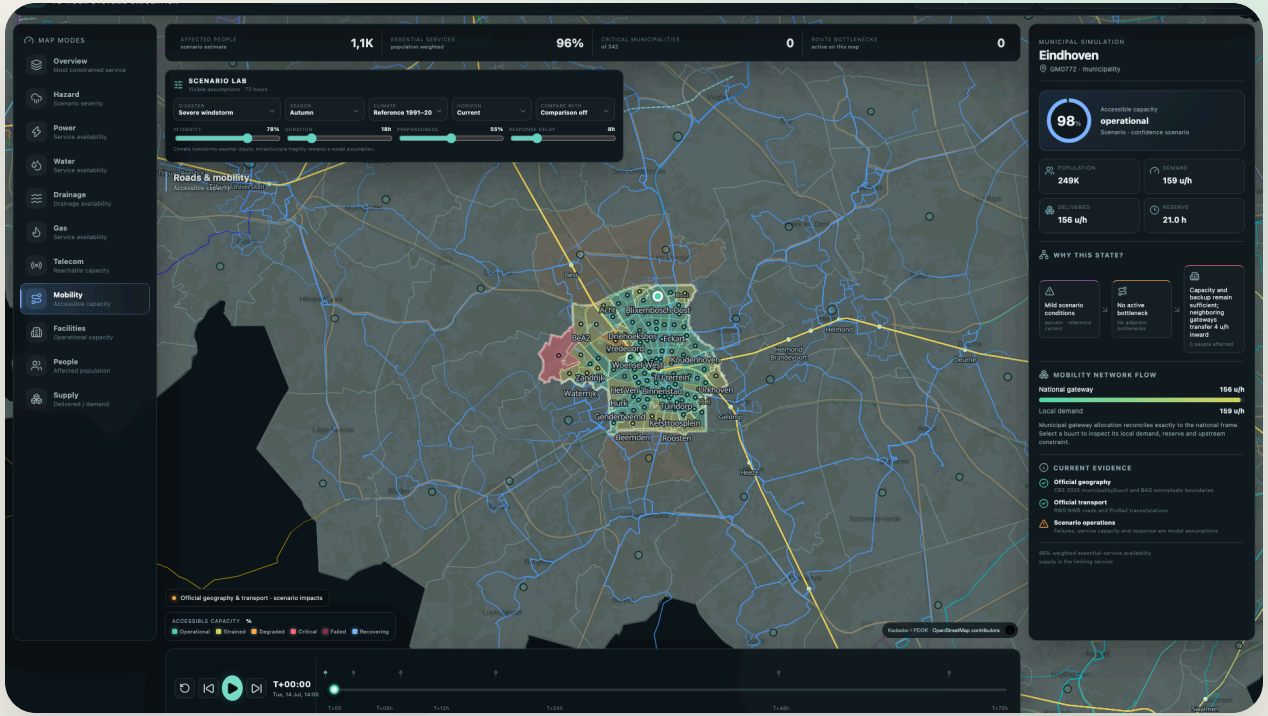
ONE MODEL, THREE SCALES

Zoom in without changing the story.

01 **Netherlands**
National balancing, affected-region footprint and 342 municipalities.

02 **Municipality**
Demand, delivery, reserve and one stable causal trace.

03 **CBS buurt**
Local impact allocation using population and vulnerability context.



Local allocation redistributes municipal inflow; it never creates supply.

PUBLIC DATA, EXPLICIT ROLES

Not every visible layer drives the model.

PLACE + PEOPLE	CBS, PDOK, BAG and BRT TOP10NL
MOBILITY	NWB, ProRail, OVapi, NDOV, NDW and FIS
HAZARDS	KNMI, RWS Waterinfo, LIWO, AHN4, RIVM and Risicokaart
SERVICES	Grid operators, HWH, DAMO, GWSW, RDI and ENTSOG
FACILITIES	KVK public services, RIVM acute care and DUO education

CAUSAL

Scenario controls and model inputs directly affect the simulation.

STRUCTURAL

Geography and audited connectivity allocate state and supported flows.

CONTEXT

Timestamped official observations orient the user, but remain non-causal by default.

UNKNOWN

Missing support stays visible. It never silently becomes zero impact or open access.

Every refresh preserves source URL, edition, licence, retrieval time and caveats.

HONEST SEMANTICS

Observed is not the same as simulated.

Official context

Boundaries, assets, provider areas and registers.

Where things are

Observed context

Radar, stations, incidents and current measurements.

What was observed

Scenario state

Synthetic hazard, availability, demand and recovery.

What the model explores

Unknown / NoData

Missing coverage, horizon, permission or semantics.

What must not be inferred

NO DATA ≠ ZERO HAZARD ≠ OPEN ROUTE ≠ HEALTHY SERVICE

WHY THIS STATE?

The explanation is part of the engine.

- **BASELINE RESILIENCE** — starting state
- **DIRECT HAZARD** — exact stage delta
- **RESTORATION** — repair progress
- **DEPENDENCY** — power / mobility
- **INTERVENTION** — policy effect
- **NETWORK TRANSFER** — supported route effect



CONTROLLED COMPARISON

Ten starting points. One lever at a time.

- SEVERE WINDSTORM
- URBAN CLOUDBURST
- RIVER HIGH WATER
- COASTAL SURGE
- HEATWAVE + DROUGHT
- WINTER STORM
- CYBERATTACK
- GRONINGEN EARTHQUAKE
- WILDFIRE + SMOKE
- COMPOUND CASCADE

- Preparedness
- Response delay
- Backup endurance
- Repair capacity
- Climate path
- Season + demand
- Network capacity
- Household readiness

Hold the event constant → change one assumption → inspect one downstream effect.

USEFUL BECAUSE THE LIMITS STAY VISIBLE

Designed to fail safely.

245

OF 342 MUNICIPALITIES AUDITED

Unsupported local pathways are withheld instead of rendered as confident detail.

WHAT IT DOES WELL

- Compare assumptions and policies
- Reveal time-dependent cascades
- Keep national and local views coherent
- Expose drivers and counterfactuals
- Preserve provenance and uncertainty

WHAT IT DOES NOT CLAIM

- A calibrated digital twin or forecast
- Live utility outage telemetry
- Physical utility network topology
- Observed affected-person counts
- Real-world units for route capacity

Audit snapshot generated 3 September 2026. Human review remains necessary.

FOUR-MINUTE WALKTHROUGH

Tell one causal story.

01

Start national

Choose Compound cascade and keep the affected region visible.

02

Move through time

Scrub from T+0 to T+24 and watch the headline state change.

03

Follow a dependency

Move from Power to Telecom, or Mobility to Supply.

04

Zoom to Eindhoven

Compare local demand, delivery and reserve.

05

Change one lever

Increase backup or reduce response delay.

06

Explain the result

Open “Why this state?” and finish on the causal trace.

Change one assumption. Show one downstream effect.

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THE INVITATION

See the cascade. Question the assumptions. Improve the response.

github.com/Rik-Stokmans/DisasterSimulator

rikstokmans.com/warroom/projects/disasterSimulator

